

STAT 525

Chapter 3
Diagnostics and Remedial Measures

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“Model Health Check”

Diagnostics

x Correctness

- Procedures to determine appropriateness of the model and check assumptions used in the standard inference
- If there are violations, inferences and modeling may not be reasonable thereby resulting in faulty conclusions ↓ Reproducibility
- Always check before any delivery of statistical conclusions
- Procedures involve both graphical methods and formal statistical tests ? Comparable

Diagnostics for X and Y

① Histogram (Bell-Shape $\rightarrow$ Normal Distribution)

② Scatterplot of Y vs X common diagnostic

- Fit smooth curve $\rightarrow I=SM\#\#$ (e.g., $I=SM70$ in slide 1-5)

- Is a linear trend ^{visually} reasonable?

- Any unusual/influential (X, Y) observations?

- Can also look at distribution of X alone

- Unusual or ^{outliers} outlying values?

- Does X have pattern over time (order collected)? ^{Violate certain spacial relationship}

- Possibly useful information of the distribution of X

- Recall model **does not** require $X \sim \text{Normal}$

- If Y depends on X , looking at Y alone may be deceiving (i.e., mixture of normal dists)

analyze one variable at a time

PROC UNIVARIATE in SAS

Evaluate distributional assumptions

- Provides numerous graphical and numerical summaries
 - Mean, median
 - Variance, std dev, range, IQR
 - Skewness, kurtosis
 - Tests for normality
 - Histograms
 - Box plots
 - QQ plots
 - Stem-and-leaf plots

Example: Grade Point Average

```
options nocenter; /* output layout: not centerized */
options colors=(none); /* graphics display: black/white */
```

```
data a1;
    infile 'U:\.www\datasets525\CH01PR19.txt';
    input grade_point test_score;
```

```
/* Legacy Line printer plots: ("ods graphics off;")
    stem-and-leaf (or horizontal bar chart)
    box plot, normal probability plot */
```

```
/* Graphics display plots: histogram, boxplot, normal qqplot*/
```

```
proc univariate data=a1 plot;
```

```
var test_score;
```

```
qqplot test_score / normal (L=1 mu=est sigma=est);
```

```
histogram test_score / kernel(L=2) normal;
```

```
run; quit;
```

line type (1: solid line)

estimate

type out the individual plot if you want to customize, or better display.

KDE

μ

mean

σ

standard deviation

(2: dash line)

solid line of normal distribution as reference

estimate mean/std dev instead of assuming 0 and 1.

The UNIVARIATE Procedure

Variable: test_score

Moments

N	120	Sum Weights	120
Mean	24.725	Sum Observations	2967
Std Deviation	4.47206549	Variance $\hat{=} (\text{Std dev})^2$	19.9993697 $\hat{=} 20$
Skewness $\hat{=} 0$ good	-0.1363553	Kurtosis $(4.47)^2$	-0.5596968
Uncorrected SS	75739	Corrected SS	2379.925
Coeff Variation	18.0872214 moderate	Std Error Mean	0.40824186

Basic Statistical Measures

Location

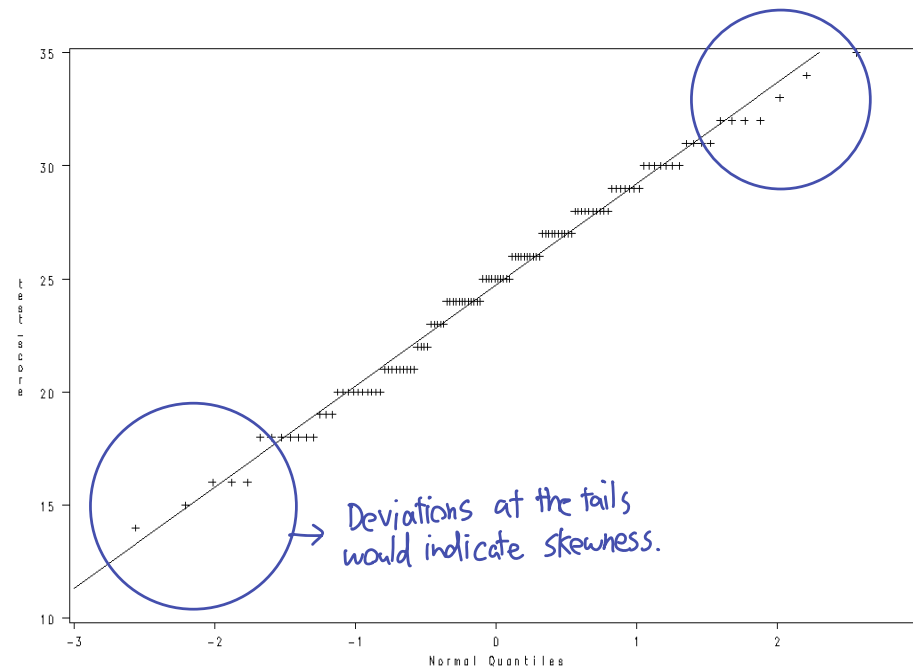
Variability

Mean	24.72500
Median	25.00000
Mode	24.00000

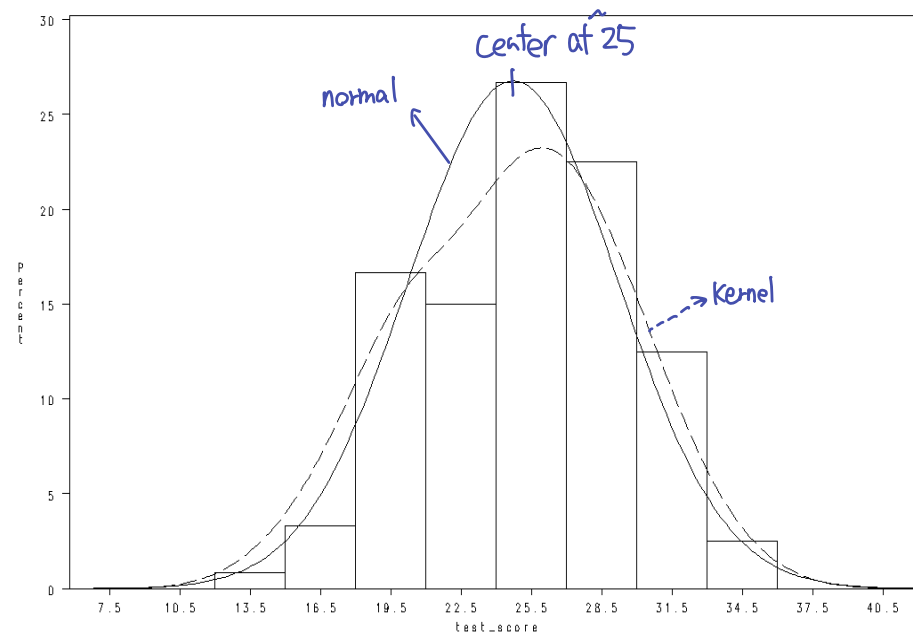
Std Deviation	4.47207
Variance	19.99937
Range	21.00000
Interquartile Range	7.00000

Mean $\hat{=}$ Median $\hat{=}$ Mode

• Distribution looks symmetrical
...



✓ points follow along the reference line
 → follow Normal Distribution



✓ Normal curve and kernel curve have similar overlay
 → follow Normal Distribution

Upper – QQ Plot

Lower – Histogram

Diagnostics for Residuals

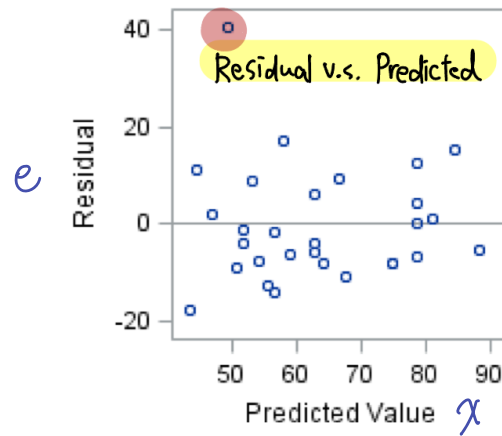
- If model is appropriate, residuals should reflect assumptions on error terms

$$\varepsilon_i \sim \text{i.i.d. } N(0, \sigma^2)$$

- Recall properties of residuals
 - $\sum e_i = 0 \rightarrow$ Mean is zero
 - $\sum (e_i - \bar{e})^2 = \text{SSE} \rightarrow$ Variance is MSE
 - e_i 's not independent (derived from same fitted regression line)
 - When sample size large, the dependency can basically be ignored
 - Residuals should look random and normal-like
 - ☒ Show patterns (x nonlinearity, heteroscedasticity, correlation... etc)

- Questions addressed by diagnostics

- Is the relationship linear? *Linearity* *If not, then we need Transformations/Nonlinear Models*
- Does the variance depend on X ? *Constance Variance (Homoscedasticity)*
- Are there outliers? *Outliers*
- Are error terms not independent? *Independence*
- Are the errors normal? *Normality*
- Can other predictors be helpful? *Model Specification*



Random scatter $\rightarrow$ good.

Residual Plots

- Plot e vs X can assess most questions
- Get same info from plot of e vs $\hat{Y}$ because X and $\hat{Y}$ linearly related
- Other plots include e vs time/order, a histogram or QQplot of e , and e vs other predictor variables
- See pages 102-113 for examples
- Plots are usually enough for identifying gross violations of assumptions (since inferences are quite robust)

Example: Toluca Company

```
data a1;
  infile 'U:\.www\datasets525\CH01TA01.txt';
  input lotsize workhrs;
  seq = _n_;

/* Default output of proc reg includes plot of residuals vs predictor */
proc reg data=a1;
  model workhrs=lotsize;
  output out=a2 r=resid;

proc gplot data=a2;
  plot resid*lotsize;
  plot resid*seq;
run;
```

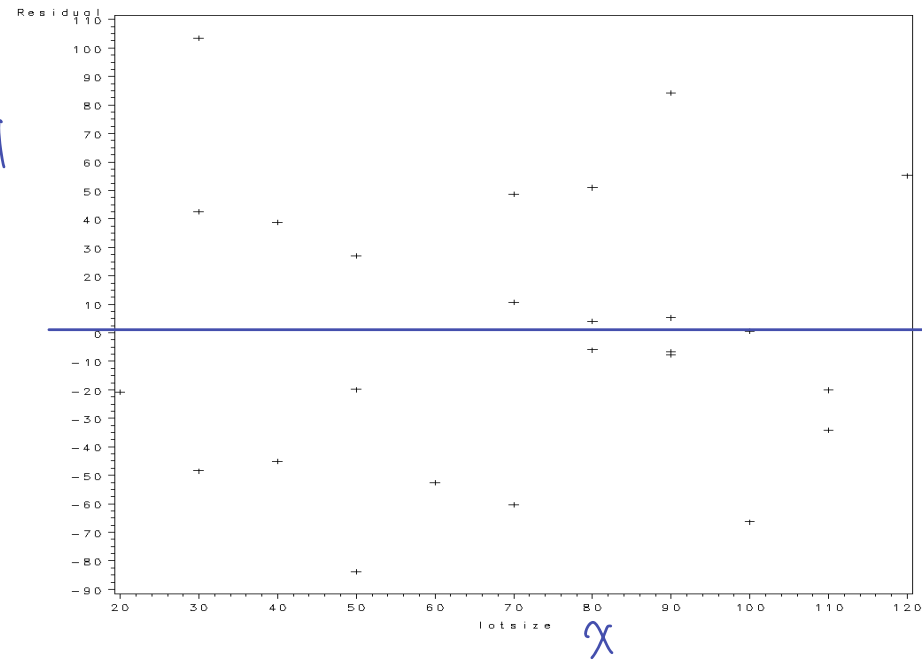
Residuals

↓ checks if residuals depend on time/order

```
/* Line type: L=1 for solid line; L=2 for dashed line */
proc univariate data=a2 plot normal;
  var resid;
  histogram resid / normal kernel(L=2);
  qqplot resid / normal (L=1 mu=est sigma=est);
run;
```

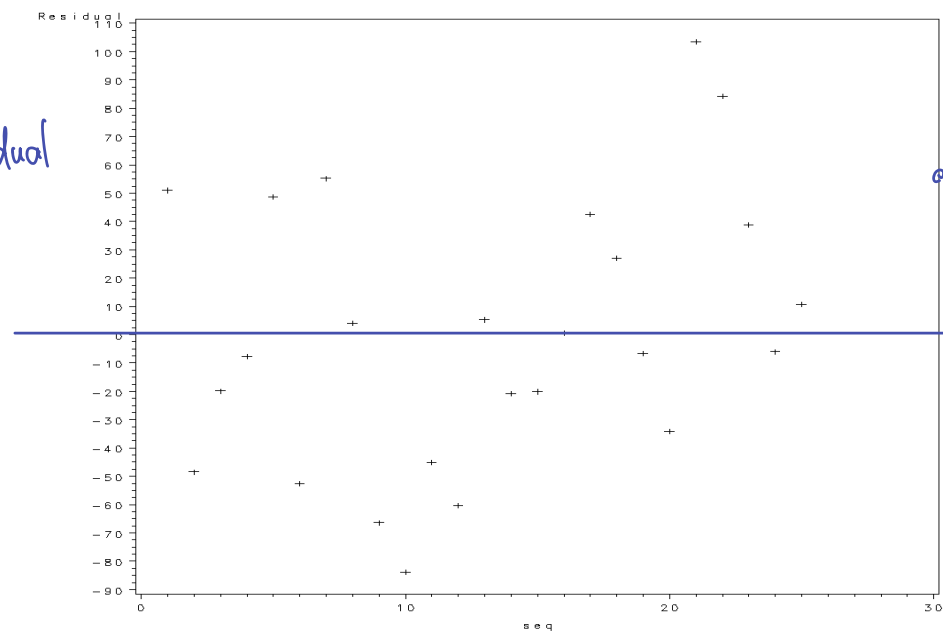
Normality check

Residual



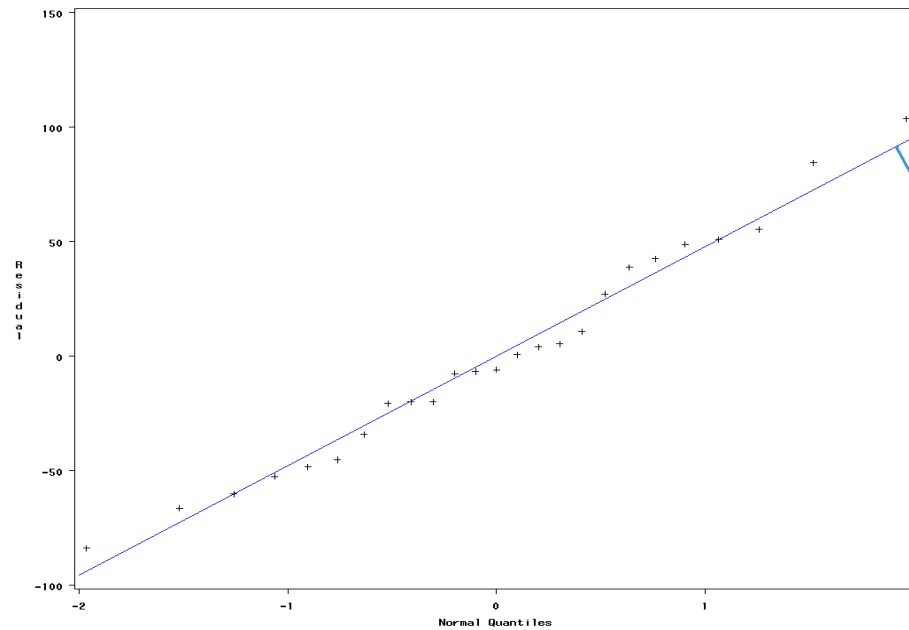
✓ Random scatter

Residual

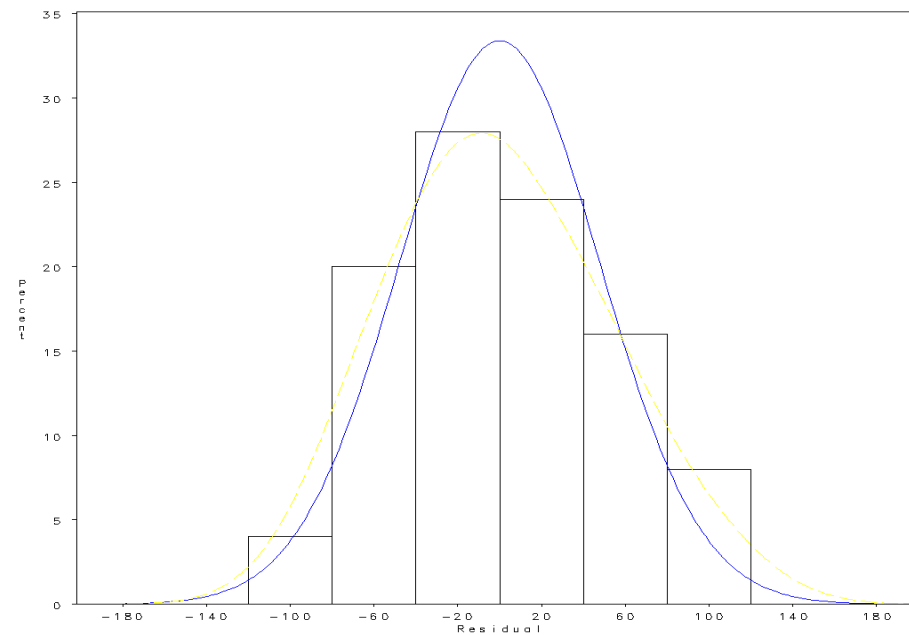


• Check independence over time/order

If:
 Curved pattern,
 S-Shape
 Systematic
 Deviations



Theoretical normal quantiles



Upper – QQ Plot

Lower – Histogram

Tests for Normality

Correlation close to 1 → look normal



- Test based on the correlation between the residuals and their expected values under normality proposed on page 115
- Requires table of critical values
- SAS provides four normality tests

```
proc univariate normal;
```

```
var resid;
```

- Shapiro-Wilk most commonly used

Example: Plasma Level (p. 132)

The UNIVARIATE Procedure

Variable: resid (Residual)

Tests for Normality					
Test	--Statistic--		-----p Value-----		
Shapiro-Wilk	W	0.839026	Pr < W	0.0011	→ Reject normality strongly
Kolmogorov-Smirnov	D	0.167483	Pr > D	0.0703	→ Fail to Reject normality
Cramer-von Mises	W^2 W-Sq	0.137723	Pr > W-Sq	0.0335	
Anderson-Darling	A^2 A-Sq	0.95431	Pr > A-Sq	0.0145	



◦ Different tests may give different results as each emphasize different aspects of distribution.

3/4 show residuals are not normal, but no need to panic:

- Look at plots to see if appropriate (graphical evidence)
- Depend on sample size

Beyond Normality Checks

Other Formal Tests

- Durbin-Watson test for correlated errors (assuming AR(1) for errors as in Chapter 12)
 ◦ time-series
◦ ordered data
- Modified Levene / Brown-Forsythe test for constant variance (Chapter 18)
- Breusch-Pagan test for constant variance

- *Priority* Plots vs Tests

Plots are more likely to suggest a remedy. Also, test results are very dependent on n . With a large enough sample size, we can reject most null hypotheses even if the deviation is slight

sample size

transformations
weighting
↑

large n ↑,
small deviations look significant

Lack of Fit Test

- More formal approach to fitting a smooth curve through the observations

replicate Y on the same level of X .

- Requires repeat observations of Y at one or more levels of X

- Assumes $Y|X \stackrel{ind}{\sim} N(\mu(X), \sigma^2)$

- $H_0 : \mu(X) = \beta_0 + \beta_1 X$ H_0 : linear

$$H_a : \mu(X) \neq \beta_0 + \beta_1 X \quad \text{H}_A: \text{not linear}$$

- Will use Allows separate mean at each X level (y_i) full/reduced model framework

↓
assumes linear regression

- ✧ Is the variation around the fitted regression line larger than the variation within replicate groups at each X ?

If yes $\rightarrow$ evidence of lack of fit (linear model may be wrong).

compare full/reduced SSEs.

- Notation

- Define X levels as $X_1, X_2, \dots, X_c$
- There are n_j replicates at level X_j ($\sum n_j = n$)
- Y_{ij} is the i^{th} replicate at X_j

- Full Model: $Y_{ij} = \mu_j + \varepsilon_{ij}$ no linearity

- No assumption on association : $E(Y_{ij}) = \mu_j$
- There are c parameters
- $\hat{\mu}_j = \bar{Y}_{.j}$ and $s^2 = \sum \sum (Y_{ij} - \hat{\mu}_j)^2 / (n - c)$

most flexible, fits each group mean

- Reduced Model: $Y_{ij} = \beta_0 + \beta_1 X_j + \varepsilon_{ij}$ assume linearity

- Linear association
- There are 2 parameters
- $s^2 = \sum \sum (Y_{ij} - \hat{Y}_j)^2 / (n - 2)$

assumes a linear structure

- $SSE(F) = \sum \sum (Y_{ij} - \hat{\mu}_j)^2$ *group mean smaller SSE*
- $SSE(R) = \sum \sum (Y_{ij} - \hat{Y}_j)^2$ *fitted larger SSE*

$$F^* = \frac{(SSE(R) - SSE(F)) / ((n - 2) - (n - c))}{SSE(F) / (n - c) \rightarrow MSE}$$

- Is *Reduced* variation about the regression line substantially bigger than variation at specific level of X ? *Full* *Meaning of Lack of Fit: linear model is too simple. > Reject H₀*
- Approximate test can be done by grouping similar X values together

Example: Plasma Level (p. 132)

```
/* Analysis of Variance - Reduced Model */
```

```
proc reg; → Regression
```

```
model lplasma=age;
```

```
run;
```

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	1	0.52308	0.52308	134.03	<.0001
Error	23	0.08976	0.00390		
Corrected Total	24	0.61284			

```
-----
```

```
/* Analysis of Variance - Full Model */
```

```
proc glm;
```

```
class age;
```

```
model lplasma=age;
```

```
run;
```

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	0.53854	0.13463	36.24	<.0001
Error	20	0.07430	0.00372		
Corrected Total	24	0.61284			

$$F^* = \frac{(.08976 - .07430)/(23 - 20)}{.00372} = 1.387$$

↓

$$\text{P-value} = 0.2757$$

✓ Fail to Reject, no evidence of Lack of Fit. → Adequate

Remedies

- Nonlinear relationship
 - Transform X/Y or add additional predictors
 - Nonlinear regression
- Nonconstant variance
 - Transform Y ($\log, \sqrt{Y}$)
 - Weighted least squares
- Nonnormal errors
 - Transform Y
 - Generalized Linear model
- Nonindependence
 - Allow correlated errors
 - Work with first differences (*time-series*)

Nonlinear Relationships

- Transformation of explanatory variables (try-and-error process)

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i \quad (\text{convex scatterplot}) \quad \text{Adding quadratic term}$$
$$Y_i = \beta_0 + \beta_1 \log(X_i) + \varepsilon_i \quad (\text{concave scatterplot}) \quad \log \text{ transformation}$$

- Transformation of the response variable

$$Y_i = \beta_0 \exp(\beta_1 X_i) \varepsilon_i \quad \text{exponential growth}$$

↓

$$\log(Y_i) = \log(\beta_0) + \beta_1 X_i + \log(\varepsilon_i) \quad \text{Taking logs makes it linear}$$

- Have altered our assumptions about error

- Can perform nonlinear regression (PROC NLIN)
 Nonlinear regression

Nonconstant Variance/Nonnormality

- Will discuss weighted analysis in Chapter 11
- Nonconstant variance often associated with a skewed error term distribution
- A transformation of Y often remedies both violations ↴
 - Stabilize variance
 - Makes errors more normal
- Will focus on Box-Cox transformations

$$Y' = Y^\lambda$$

Box-Cox Transformation

data-driven way to decide to "best" power transform for Y , instead of guessing.

- Special cases:

$\lambda = 1 \rightarrow$ no transformation

$\lambda = .5 \rightarrow$ square root

$\lambda = 0 \rightarrow$ natural log (by definition)

- Can estimate λ using ML (Maximum Likelihood)

$$f_i = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{1}{2\sigma^2} (Y_i^\lambda - \beta_0 - \beta_1 X_i)^2 \right\} \times \text{Jacobian}$$

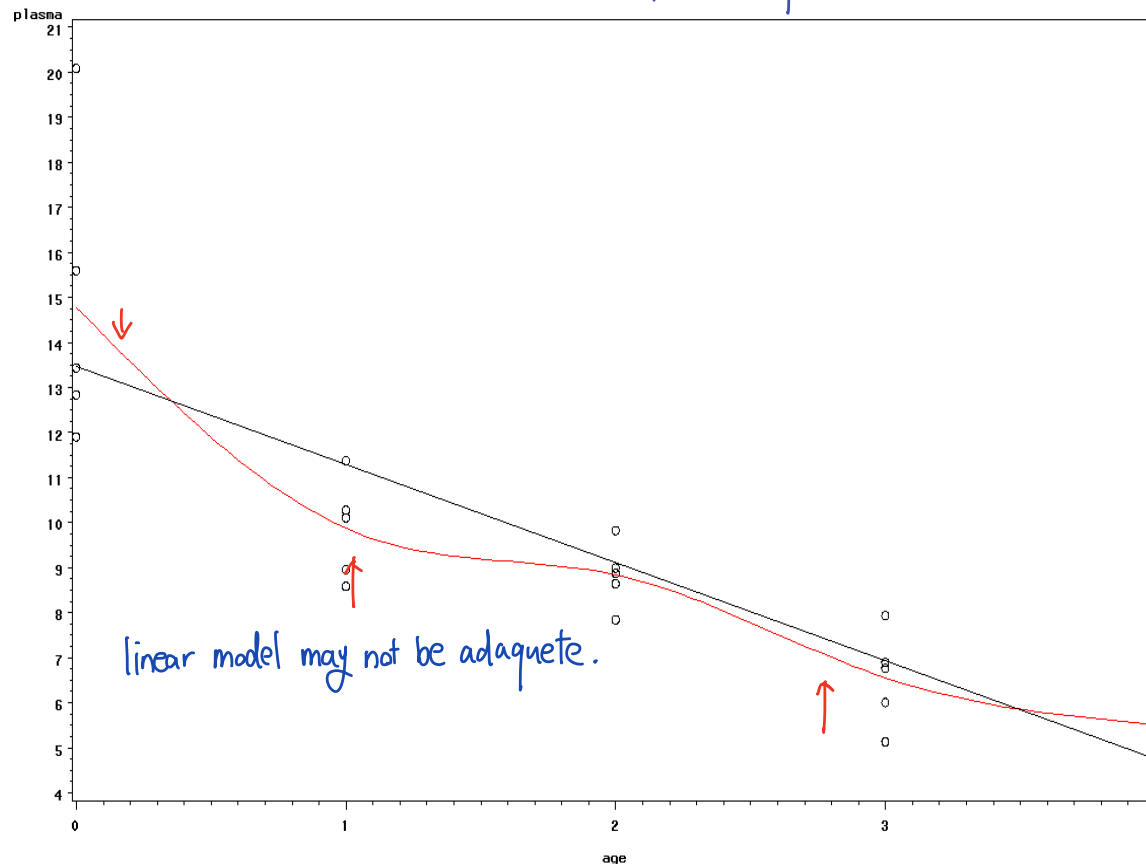
– λ_{ML} minimizes SSE of standardized Y^λ 's.

- Can also do a numerical search
- PROC TRANSREG will do this in SAS

Example: Plasma Level (p. 132)

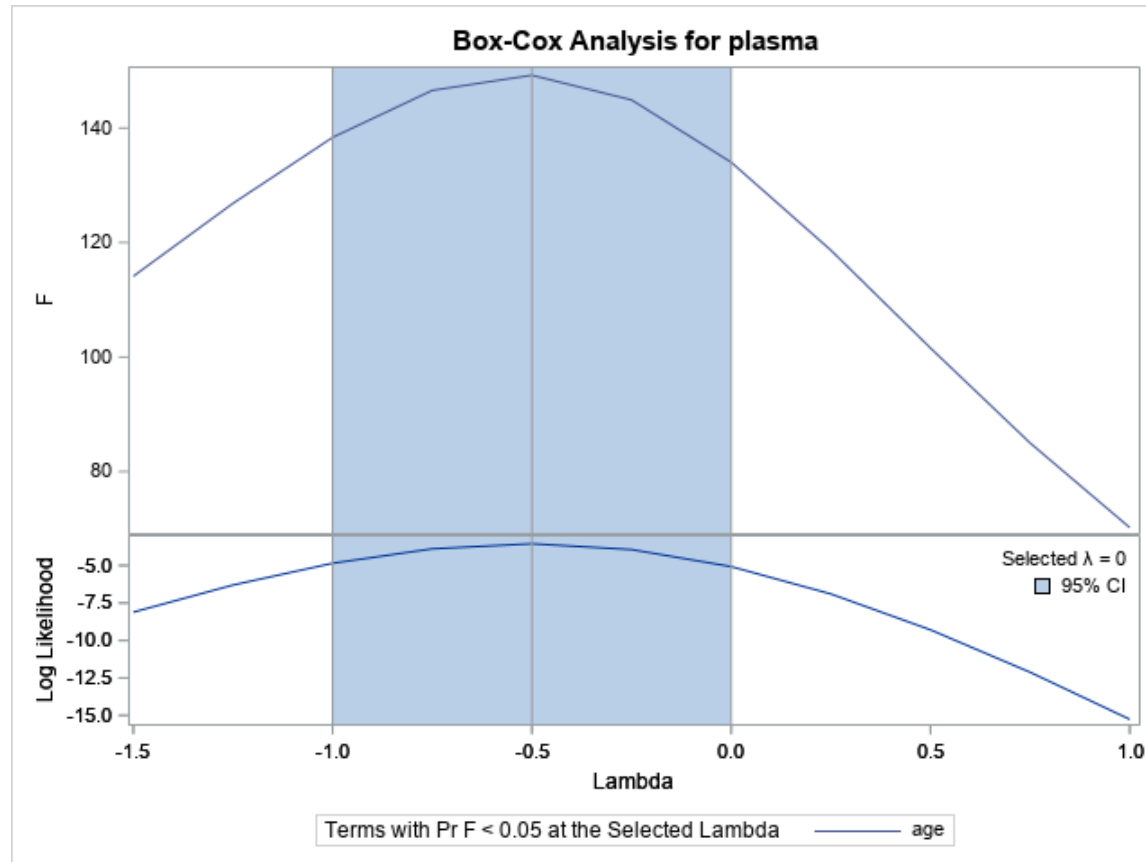
```
data a1;  
  infile 'd:\nobackup\tmp\CH03TA08.txt';  
  input age plasma lplasma;  
symbol1 v=circle i=sm50 c=red; symbol2 v=circle i=r1 c=black;  
proc gplot;  
  plot plasma*age=1 plasma*age=2/overlay; run;
```

Raw response *log-response*
Regression Line
lines on the same plot



```
proc transreg data=a1;
  model boxcox(plasma/convenient parameter = 0 lambda=-1.5 to 1 by 0.25)
    =identity(age);
run;
```

If MLE ≈ 0 , Report exactly 0.
 Shift Y values so none are non-positive
 keep age untransformed
 test a range of λ (power) values



- `parameter=c` shift all Y by c , in case that some Y 's are negative
- `convenient` option gives the most convenient value $\lambda = 0$ (i.e., log transform)

```
proc gplot;
```

```
plot lplasma*age=1 lplasma*age=2/overlay;
```

```
run; quit;
```



Chapter Review

- Diagnostics
 - Graphical methods
 - Statistical tests
- Remedies
 - Nonlinearity
 - Nonconstant variance

Lambda (λ)	Transformation Applied to Y	Name
-1	$\frac{1}{Y}$	Reciprocal
-0.5	$\frac{1}{\sqrt{Y}}$	Reciprocal square root
0	$\log(Y)$	Logarithmic
0.5	$\sqrt{Y}$	Square root
1	Y	No transformation
2	Y^2	Square